RESEARCH PROJECT OVERVIEW

Risk Monitoring for LLM Trading Agents

Detection, prediction and intervention evaluation from a market supervisor’s perspective

TARGET MARKET: Crypto perpetual futures (BTC / ETH perp)

Phase 1 Risk discovery

Phase 2 Intervention testing

Phase 3 Detection & prediction

Oct. 23, 2026

PROJECT POSITIONING

Regulators need to see risk build up, not count agents

Core problem

As LLM trading agents become more common, their collective behaviour may amplify market risk: similar models and data can push many agents to trade the same way at the same time.

What a supervisor cannot do

- Control how many agents trade
- See which model produced an order
- Inspect agents' internal structure

So the project monitors observable market conditions instead.

Final output: a multi-risk monitoring system

Input	Observable market indicators (volatility, spread, depth, order imbalance)
Detect	Risk level per dimension: Normal / Watch / Warning / Critical
Predict	Short-term probability that each risk escalates
Support	Which interventions could reduce risk under current conditions

Source: FSB (2024), The Financial Stability Implications of AI

METHOD

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Simulation is the method, not the goal

Real markets cannot tell us how risk changes with the number of LLM agents, so we build a calibrated synthetic market and run controlled experiments in it.

1 Controlled laboratory

Isolate the effect of agent density N under repeatable conditions: same scenarios, same rules, same information.

2 Counterfactual reasoning

Answer “what if N were different?” or “what if a leverage cap were in place?”, which historical data cannot answer.

3 Labelled training data

Produce step-by-step market states s_t with known risk labels R_t: the only feasible source of labels for training the detector.

Bridge to real markets

The simulator reproduces a perpetual-futures venue: a continuous order book, leverage, hourly funding settlement and an automatic liquidation engine. The detector itself still reads only indicators a trading venue can observe.

WHAT CHANGED Target market changed from US equities to crypto perpetual futures. The three reasons for simulating — control, counterfactual reasoning and labelled data — are unchanged.

REVISION · RATIONALE

NEW

Why we moved from US equities to crypto perpetual futures

Three reasons: where agents actually trade, what data lets us see, and which question is still open.

AGENTS ARE NOT THERE YET

The problem with equities

No population-scale evidence of LLM-agent trading in US equities in 2026.

Retail access is gated by broker-dealer controls: SEC Rule 15c3-5, MiFID II algo rules, ESMA supervisory briefing (Feb 2026).

We would be studying a phenomenon that has not yet formed.

THIS IS WHERE AGENTS TRADE

The evidence for perpetuals

arXiv 2609.05663 (2026) records two production LLM-agent fleets: 3,505 funded vaults and 500–599 agents trading Hyperliquid perpetuals.

~7.5M model calls, 300k on-chain actions, 14,596 trades over six months.

Median leverage 5.0x and insensitive to volatility; 11% of positions produced 62% of liquidations.

WE CAN SEE THE CHAIN

The methodological gain

Perpetual venues publish liquidation, funding-rate and open-interest streams in real time.

Equity data carries no liquidation field: forced selling stays hidden inside broker–client relationships.

Our chain — agents → leverage → forced liquidation → cascade → systemic risk — becomes observable end to end.

SCOPE NOTE What does NOT change: the detector still reads only venue-observable indicators, and US equities stay in the study as a secondary transferability check rather than being dropped.

Source: arXiv 2609.05663 (2026); SEC Rule 15c3-5; ESMA Supervisory Briefing on Algorithmic Trading (Feb 2026); FSB final AI report (Oct 2026)

NEW

What a crypto perpetual is, and why it fits our question

For readers who know equities: three mechanisms are added, and each one matters for risk.

	US equity	Crypto perpetual
Trading hours	6.5 h per day	24 / 7
Typical leverage	~2x (Reg T)	5x – 50x
When losses mount	Broker issues a margin call	Engine force-closes and logs it
Circuit breakers	Market-wide + LULD bands	None — an open policy question
Best-price protection	NBBO / Reg NMS	None; single-venue book
Market-maker duty	Yes	None; makers may walk away
Cost of holding	Margin interest	Funding rate (longs ↔ shorts)

Funding rate · the rent that keeps price honest

Longs pay shorts when price is above spot. Extreme funding means crowded leveraged longs: a free fragility signal.

Leverage + maintenance margin

5x means 100 of margin controls a 500 position. When losses eat the margin below the maintenance level, the venue closes it automatically.

Liquidation cascade · the domino effect

Forced selling pushes price down → more positions breach → more forced selling. This is what the detector must catch.

WHY THIS MATTERS Why it fits: our research question is whether agent populations create systemic risk. In perpetuals every link of that chain sits in the public data stream. In equities the most important link — forced liquidation — is invisible.

FRAMEWORK

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What goes into each phase, and what comes out

	PHASE 1 Risk discovery	PHASE 2 Intervention testing	PHASE 3 Detection & prediction
INPUT	Agent density N (control variable) Market state M: normal / stress 8 perpetual-market scenarios Leverage, funding, liquidation rules fixed	Phase 1 baseline results Intervention parameters The same 8 scenarios Paired re-adaptation runs (new)	Observable indicators s_t only: volatility, spread, depth, order-flow imbalance, funding rate, liquidation intensity, open interest, rates of change
DATA FLOW	**Vary N → run simulation → record (s_t, R_t)**	**Apply interventions → re-adapt → record**	**Train on all (s_t, R_t) → deploy with s_t only**
OUTPUT	Systemic-risk metrics as functions of N Per-step market state s_t and risk value R_t **Key: (s_t, R_t) pairs = labelled training data**	Intervention effectiveness table Short-run vs. post-adaptation effect **Key: post-intervention data enrich training**	Risk level per dimension + forecast Cross-market transferability check **Key: N is not needed at deployment**

WHAT CHANGED Scenarios reduced 20 → 8 (factorial design). Added funding rate, liquidation intensity and open interest as observable inputs — all exist in perpetual-futures data but have no equity equivalent.

REVISED

Three risk dimensions form one causal chain

DRIVER

Behavioural synchronisation

Pairwise correlation of order directions (simulation); same-direction order-flow persistence (real data)

→

MEDIATOR

Liquidity fragility + forced liquidation

Order-book depth decline rate; bid–ask spread widening speed; liquidation-cascade intensity (new)

→

OUTCOME

Systemic risk

Volatility amplification; max drawdown; joint drawdown probability; risk-contribution concentration

Why these three: academic and regulatory basis

FSB (2024)

names market correlations as one of four AI vulnerabilities that could raise systemic risk: common models and data "could lead to increased correlations".

CFTC–SEC (2010) & BIS (2026)

describe the flash crash as "two liquidity crises": depth fell below 1% of its morning level. BIS (2026) flags correlated, momentum-driven AI positioning.

IMF (2024) & Bank of England (2026)

warn of faster, more volatile markets if AI strategies "respond to a shock in a similar manner". The BoE adds: frameworks were not built for autonomous agents.

WHAT CHANGED Added liquidation cascade to the mediator, and funding rate as an observable early signal. Both are measurable in perpetual-futures data; the cascade is the mechanism our research question is about.

Source: FSB (2024, 2026); CFTC–SEC (2010); IMF GFSR Oct 2024; Bank of England FSR Jul 2026; arXiv 2609.05663

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Phase 1 varies agent density under controlled conditions

Objective

Measure how each risk dimension R_k changes as the number of independent LLM agents N increases, in normal and stress markets M.

$$R_k = f_k(N, M) + \varepsilon$$

Key variables

Independent	N ∈ {5, 20, 50, 80, 100}
Market state	Normal vs. stress
Repeats	Stratified 6 / 6 / 12 / 12 / 12
Held constant	Role mix (25% each), LLM backbone, information, leverage cap, funding rule, initial capital

Data generation

- Synthetic perpetual market calibrated to real Level-2 order-book data
- 8 fixed environment scenarios shared across all conditions (was 20)
- 100 time steps per run; step length fixed during calibration
- Stratified repeats: 6 / 6 / 12 / 12 / 12 across density

Design note
Stratified repeats put effort where variance is highest — near the critical density — and cut LLM calls by about 74% with no loss of resolution.

WHAT CHANGED Calibration source changed to crypto Level-2 data. Scenarios 20 → 8; uniform 20 repeats → stratified 6/6/12/12/12. Estimated LLM calls fall from ~30.6M to ~7.9M, bringing the project inside a student budget.

PHASE 1 · RISK DISCOVERY

Phase 1 turns simulation runs into data-driven risk levels

1 Measure

- Outcome: volatility amplification, max drawdown, joint drawdown probability, risk-contribution Gini
- Mechanism: order-direction correlation, depth decline rate, spread widening speed

2 Find thresholds

- Apply change-point detection (PELT / CUSUM) to each risk series
- Locate where the risk structure shifts
- Thresholds come from the data, not chosen by hand

3 Deliver

- Risk–density curves, normal vs. stress
- Critical density thresholds
- Initial level boundaries
- Composite MRI curve
- Mechanism analysis

Composite Market Risk Index $MRI_t = \Sigma_k w_k \cdot \tilde{R}_{k,t}$ $\tilde{R}$ = normalised risk score, $\Sigma w_k = 1$; weights set and justified in Phase 1

Four risk levels per dimension

Normal
Stable conditions

Watch
Material deterioration

Warning
High chance of escalation

Critical
Severe event under way or imminent

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Phase 2 tests existing controls, plus perpetual-specific ones

Standard venue controls (retained)

Measure	Parameter	Target risk
Order size cap	Max quantity per order	Systemic, liquidity
Order frequency limit	Min order interval	Synchronisation
Position concentration	Max % of market value	Systemic
Directional exposure cap	Max one-sided exposure	Systemic, sync.
Circuit breaker / pause	Price-move threshold	Systemic, liquidity
Pre-trade check	Reject breaching orders	All, at entry

Real-rule basis

Pre-trade price and size controls are already required of algorithmic traders under MiFID II RTS 6 and of US broker-dealers under SEC Rule 15c3-5.

Perpetual-specific controls (NEW)

Measure	Parameter	Target risk
Leverage cap	Max leverage per account	Cascade
Funding-rate clamp	Cap on |funding|	Crowded leverage
Maintenance-margin tier	Margin by position size	Cascade
Liquidation auction	Auction vs. market order	Cascade, liq.
ADL redesign	How winners are cut	Extreme loss

Why this block is new

No 2025–2026 study tests circuit breakers, transaction taxes or position limits against LLM-agent-driven instability. Perpetual-specific controls are an empty cell in the literature — and a live policy question for venues today.

WHAT CHANGED Added five perpetual-specific interventions targeting the liquidation cascade. The original six controls are retained, so equity-market comparability is preserved.

Source: MiFID II RTS 6; SEC Rule 15c3-5; Cboe LULD FAQ; ESMA algorithmic trading guidelines; venue rulebooks (Hyperliquid, Binance, OKX)

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Each control is tested alone, then re-tested after agents adapt

Experimental design

1. Baseline: Phase 1 runs with no intervention
2. Apply each intervention on its own
3. **Let the agent population re-adapt, then measure again (NEW)**
4. Test 2–3 promising combinations
5. Reuse the same scenarios and random seeds
6. Record risk, liquidity side effects and trigger frequency

Interaction between two controls

$E_i = (R_o - R_i) / R_o$ risk reduction of control i

$I_{ij} = E_{ij} - (E_i + E_j)$

$I_{ij} > 0$	**Synergy: better together**
$I_{ij} \approx 0$	Additive / redundant
$I_{ij} < 0$	Conflict: overlap or offset

Evaluation metrics

- Risk reduction vs. baseline, per dimension
- Level change (e.g. Warning → Watch)
- Liquidity cost and recovery speed
- **Short-run effect vs. post-adaptation effect (NEW)**

WHAT CHANGED Added step 3, the paired re-adaptation protocol. Measuring only the immediate effect of a control overstates it, because agents re-optimise around new rules — see the next slide.

METHOD · NEW PROTOCOL

NEW

Effective on day one? Measure again after the agents adapt.

Paired re-adaptation protocol, after arXiv 2610.18158 (2026).

The two-phase protocol

Phase A — short run

Apply the control and run 100 steps. Record risk. This is the number most studies report — and it is usually the flattering one.

Phase B — after adaptation

Keep the control in place and run 500–1,000 further steps, or re-train agent policies under the new rule, until behaviour stabilises. Record risk again.

Only Phase B is the equilibrium effect. Report both.

What we report

For every intervention: a short-run effect and a post-adaptation effect. If the two diverge, the control is being routed around — which is itself a finding a supervisor needs.

WHY THIS MATTERS Why this is a contribution: no 2025–2026 study evaluates circuit breakers, taxes or position limits against LLM-agent-driven instability — and none measures the effect after agents adapt. That cell is empty.

Source: arXiv 2610.18158, “When and How Order-Flow Interventions Work After Dealers Adapt” (2026)

PHASE 2 · OUTPUT

Phase 2 maps market states to interventions

Template only. Cells are filled with measured results once the experiments are run; no values are assumed in advance.

Market state	Density	Intervention	Systemic risk Δ	Synchronisation Δ	Liquidity cost	Recommendation
Normal	Low	None	Baseline	Baseline	None	No action
Stress	Medium	Directional cap	*measured*	*measured*	*measured*	*measured*
Stress	High	Circuit breaker + cap	*measured*	*measured*	*measured*	*measured*

Interaction summary (template)

Intervention pair	Combined effect	Sum of individual effects	Interaction I_{ij}	Interpretation
Size cap + frequency limit	*measured*	*measured*	*measured*	*measured*
Frequency limit + circuit breaker	*measured*	*measured*	*measured*	*measured*

PHASE 3 · DETECTION & PREDICTION

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Phase 3 builds a detector that only uses what the market shows

Inputs s_t	→ Model	→ Outputs
Rolling volatility σ Spread / mid-price ℓ Top-of-book and book depth Order-flow imbalance + persistence Rates of change: volatility, spread, depth ▸ **Funding rate and its change** ▸ **Liquidation intensity (count, notional)** ▸ **Open interest and mark-price basis** ▸ **Same-direction flow proxy**	Training data: Phase 1 + 2 simulated states with labelled risk levels Multi-label classifier: random forest or gradient boosting Logistic regression as a transparent benchmark Time-series features over rolling windows Calibrated probabilities; class imbalance handled	Risk level per dimension, with confidence Probability of escalation within the next k steps Risk heatmap across dimensions Early-warning indicators that best predict each risk Link to Phase 2 intervention evidence

Excluded from the detector: agent count N, agent identity, private positions, and distance from fundamental value (not observable in real markets).

WHAT CHANGED Three new inputs — funding rate, liquidation intensity and open interest — are directly observable in perpetual-futures data. The detector now sees forced selling instead of inferring it from price alone.

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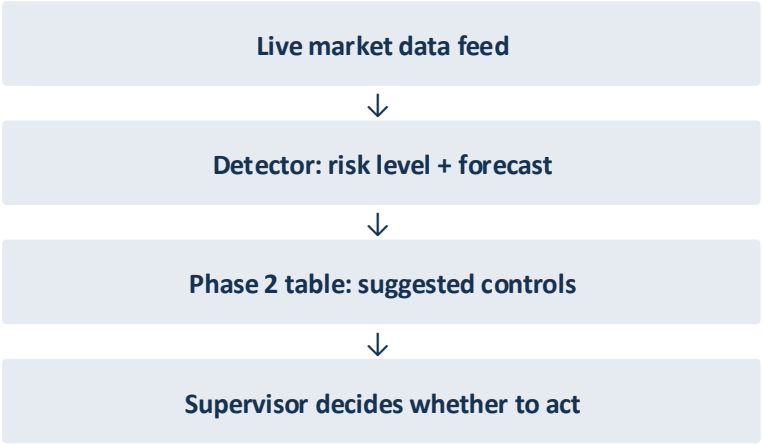
Validate on real stress events, then deploy as decision support

External validation

- Test on unseen simulation seeds and changed agent settings
- Run on real perpetual-market stress events: 19 May 2021 liquidation cascade, May 2022 Terra unwind, 5 Aug 2024 carry unwind, Apr 2025 tariff shock, 10 Oct 2025 cascade, Mar 2026
- Cross-market check on equities: Jan 2021 GameStop squeeze, Mar 2020 COVID sell-off
- Compare simulated vs. real stylized facts: volatility clustering, fat tails, order-flow autocorrelation
- Report the performance drop from simulated to real data

Limit
Real crises test whether the detector recognises stress; they cannot show that LLM agents caused it. Event-sample size is the binding limit: free perpetual history covers about one year, so we buy specific crisis windows.

Deployment form



A module for existing venue risk dashboards. It supports decisions; it never intervenes automatically.

WHAT CHANGED Validation events moved from 2010 / 2020 / 2022 equity episodes to **six perpetual-market stress events**. Two equity episodes are kept as a cross-market check on whether the framework transfers.

Source: venue liquidation and funding records; Cboe VIX history; arXiv 2609.05663 (2026)

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Which data exists now, and which features we will use

Feature in the detector	Data it needs	Where to get it	Status
A · Core: standard trade and quote data			
Returns, rolling volatility σ, volume	Trades (price, size, time)	CryptoHFTData; venue WebSocket	Use
Relative spread ℓ = spread / mid	Best bid and ask quotes	CryptoHFTData; venue WebSocket	Use
Top-of-book depth; flow imbalance	Quoted sizes; signed trades	CryptoHFTData; venue WebSocket	Use
B · Extended: needs Level-2 order-book data			
Multi-level depth; depth decline speed	Book snapshots by price level	CryptoHFTData; Tardis (1st/mo)	Use
Cancel-to-trade ratio; order-arrival rate	Order messages (add, cancel, execute)	0xArchive (paid, event windows only)	Use on events
C · Newly observable in perpetuals (new)			
Funding rate and its rate of change	Funding settlement stream	Hyperliquid info API; venue WebSocket	Use — new
Liquidation intensity and cascade depth	Liquidation event stream	Venue WebSocket; 0xArchive	Use — new
Open interest and mark-price basis	Venue state snapshots	Venue REST API	Use — new

WHAT CHANGED Block C is new: funding rate, liquidation intensity and open interest were previously simulation-only quantities. In perpetual-futures data they are published streams, so the detector can observe forced selling directly.

Source: CryptoHFTData; Hyperliquid API docs; Tardis.dev; 0xArchive; Binance Vision

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Where each dataset fits in our plan

Source	What it contains	Access / cost	Our use
CryptoHFTData	L2 book Parquet: Hyperliquid, Binance, OKX, Bybit	Free, API key	Primary calibration, from Sep 2025
Hyperliquid WS (self-recorded)	Live L2 book, trades, funding, liquidations	Free	Primary from Oct 2026 (~6 months)
Binance bookDepth	Depth by ±% band, 30 s, from Jan 2023	Free	Depth panel; stress-day screen
Tardis.dev	Incremental L2 across 69 venues, from 2019	Free on the 1st of each month	Multi-year calibration sample
0xArchive	Level-2 / Level-4 order data for Hyperliquid	~US$3 / GB	Buy 6–8 crisis windows only
SEC Market Structure Data	Daily cancels, trades, quote life; 2012–2026	Free	Equity check; benchmark
LOBSTER	Nasdaq Level-3 rebuilt from TotalView-ITCH	Student credits, 0.1 LDC/MB	Optional equity comparison
FRED	Interest rates, inflation and macro series	Free	Scenario generation

Rule A feature enters the detector only if the same field exists in both the simulator and the chosen dataset. Removed: SEC MIDAS (page retired), BitMEX S3 (exchange closed Sep 2026), WRDS TAQ (NBBO only), Databento equities (starts 2018).

WHAT CHANGED Source list replaced. Equity sources were paid, slow or depth-free; perpetual sources are free, complete and carry liquidation and funding streams. Estimated cash cost falls to roughly US$20–60.

Source: CryptoHFTData; Hyperliquid; Binance Vision; Tardis.dev; 0xArchive; SEC Market Structure Data; LOBSTER

DATA PLAN

NEW

Top-down: what each stage needs; bottom-up: what we can get

The gap is narrow, and it closes for roughly US$20–60.

Stage	What it needs	How much	What exists free	Gap → fix
G1 Agent behaviour	Real agent statistics to set behaviour	Published figures	arXiv 2609.05663: leverage, liquidation concentration, win rate	None — used directly
G2 Calibration	Representative normal-period Level-2 data	40–60 symbol-days across 5–8 years	CryptoHFTData (2025-09→), Tardis 1st-of-month (2019→), Binance bookDepth (2023→)	None — stratified sampling covers it
G3 Validation	Real stress events	6–8 events	Free history covers ~2–3 events (about one year)	Buy specific crisis windows (~US$3/GB) to reach 6–8
G4 Interventions	Counterfactual runs under each control	Simulation only	Needs no new data — the calibrated simulator supplies it	None
Equity check	Ten-year continuity	Daily aggregates + a small L2 sample	SEC Market Structure Data (2012–2026, free)	LOBSTER optional

REVISED

A deliberately narrow set-up keeps the study feasible

Agent dimension	Assumption
Trading	Market / limit orders; one decision per step; leverage capped (default 5x, from observed agent fleets)
Styles	Value, momentum, market-maker, contrarian; fixed 25% each
Risk appetite	Value: moderate; momentum: low; market maker: high; contrarian: moderate
Heterogeneity	Role prompts (main); information subsets and model diversity as extensions
Information	Public only, plus funding rate and liquidation feed; no private signals
Autonomy	No researcher intervention during a run

Feasibility

- 5 density levels
- 100 steps; 8 scenarios
- Stratified repeats 6 / 6 / 12 / 12 / 12
- Cache + batch API + structured output
- Open-source scans; strong model on 4%
- Phase 3 reuses Phase 1 + 2 data

Out of scope

- Multi-asset portfolios; private info
- Comparing LLM capabilities
- Spot crypto, on-chain MEV, DEX splits
- Automatic intervention; legal enforcement
- Collusion and manipulation (ext.)

WHAT CHANGED Market-mechanism assumptions now describe a perpetual venue (leverage cap and funding feed added). Token-saving measures moved from a separate plan into the scope of the study itself.

REVISED

From worrying about AI risk to detecting it and acting on it

Discover

Phase 1 shows when systemic risk becomes dangerous as density rises, and explains why through synchronisation, liquidity loss and forced liquidation.

Risk–density curves · risk levels

→

Define

Phase 2 shows which controls reduce each risk, how pairs interact, and which effects survive once agents adapt to the new rules.

Effectiveness table · interaction analysis

→

Solve

Phase 3 delivers a tool supervisors can use to monitor market state, read early warnings and decide when to step in.

Detector · forecast · heatmap · early warnings

Also delivered

- Policy brief for regulators on monitoring and intervention
- Reproducible code repository
- Simulator calibration report
- Cross-market transferability check (US equities)
- Data-acquisition scripts and the recorded dataset

WHAT CHANGED Two deliverables added: the cross-market transferability check (keeps the work relevant to equity regulators) and the recorded dataset with its acquisition scripts.